

IntelliScrap AI - Executive Summary

Overview

IntelliScrap AI is an offline-first Progressive Web App (PWA) empowering informal waste recyclers (Baban Bola) in Northern Nigeria to identify scrap materials, detect hazards, and get fair market prices.

Problem

- Lack of safety awareness for toxic e-waste
- Exploitation by middlemen due to price opacity
- Limited connectivity and low smartphone penetration

Solution

Photo-based AI classification with honest fallbacks (heuristic analyzer, offline-first), multilingual voice guidance, USSD/SMS for feature phones, Uber H3-based collector matching, and built-in traceability.

Key Features

- Offline-first AI (ONNX + lightweight heuristic)
- Multilingual TTS (Hausa, Pidgin, English)
- USSD/SMS & feature phone support
- Smart matching via Uber H3
- Recycling hubs & compliance (EPR)
- PWA with offline sync
- Impact tracking

Impact

- Fair pricing → improved incomes
- Hazard awareness → safer working conditions
- Traceability → EPR compliance
- Market access for unconnected collectors

Status

Production-ready PWA with live scanning, pickups system, voice guidance, and UI polished for public use.